Lecture 10 (sections 4.5.1-4.5.4)

Universal quantum gates

- Recall, in *classical computation* a Toffoli gate

$$\{c_1, c_2, t\} \xrightarrow{Toffoli} \{c_1, c_2, t \oplus c_1 c_2\}$$

  is universal

- In *quantum computation* there is a also universal set of gates:
  - Any quantum computations is a $2^n \times 2^n$ unitary transformation U on $n-$qubits,

$$|x_1 x_2 \cdots x_n\rangle \xrightarrow{\text{quantum computation}} U\,|x_1 x_2 \cdots x_n\rangle$$

  - Any U can be **exactly** decomposed as a product of 2-level unitaries U_i,

$$U = \prod_{i=1}^{k \leq 2^{n-1}(2^n-1)} U_i$$

  where each U_i acts only on 2 states in 2^n-dimensional Hilbert space of $n-$qubits (NOT necessarily a single qubit!!!!!)
  - Using above, we show that arbitrary unitary is **exactly** represented as a product of single qubit operations and CNOT gates
  - Arbitrary single qubit operation can be **approximated** to any desired precision with H, T and $S = T^2$ gates

$\Longrightarrow$ Universal circuit representation is not necessarily the most efficient; we simply care in this context that *it can be done*

Two-level unitary gates are universal

- Consider a $d \times d$, to begin we set $d = 3$, unitary:

$$U = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & j \end{bmatrix}$$

A unitary preserves a norm of a state $\Longrightarrow$

$$||U\,|\psi\rangle\,||^2 = \langle\psi|\underbrace{U^+U}_{=1}|\psi\rangle = ||\,|\psi\rangle\,||^2 = 1$$

so if we choose, *e.g.,*

$$|\psi\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \Longrightarrow \qquad U\,|\psi\rangle = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \qquad \Longrightarrow \qquad |a|^2 + |b|^2 + |c|^2 = 1$$

$\Longrightarrow$ Likewise, the norm of any column of any unitary is 1

$\Longrightarrow$ Since $V \equiv U^+$ is also unitary, the same is applicable for any row of a unitary, *e.g.,*

$$|a|^2 + |d|^2 + |g|^2 = 1$$

- We now explicitly provide 2-level unitary matrices U_1, U_2, U_3, such that

$$U_3 U_2 U_1 U = I_3$$

$\Longrightarrow$

$$U = U_1^+ U_2^+ U_3^+$$

■ We construct U_1, $U_1 U_1^+ = U_1^+ U_1 = I_3$, so that $U_1 \cdot U$ has zero in the element indexed as $_{21}$ — originally b:

$$U_1 = I_3\,, \qquad \text{if} \qquad b = 0$$

$$U_1 = \begin{bmatrix} \frac{a^*}{\sqrt{|a|^2+|b|^2}} & \frac{b^*}{\sqrt{|a|^2+|b|^2}} & 0 \\ \frac{b}{\sqrt{|a|^2+|b|^2}} & \frac{-a}{\sqrt{|a|^2+|b|^2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \text{if} \qquad b \neq 0$$

$\Longrightarrow$ the first column of $U_1 \cdot U$:

$$a \cdot \begin{bmatrix} \frac{a^*}{\sqrt{|a|^2+|b|^2}} \\ \frac{b}{\sqrt{|a|^2+|b|^2}} \\ 1 \end{bmatrix} + b \cdot \begin{bmatrix} \frac{b^*}{\sqrt{|a|^2+|b|^2}} \\ \frac{-a}{\sqrt{|a|^2+|b|^2}} \\ 1 \end{bmatrix} + c \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a' \\ 0 \\ c' \end{bmatrix}$$

$\Longrightarrow$ in general,

$$U_1 U = \begin{bmatrix} a' & d' & g' \\ 0 & e' & h' \\ c' & f' & j' \end{bmatrix}$$

■ We construct U_2, $U_2 U_2^+ = U_2^+ U_2 = I_3$, so that $U_2 \cdot U_1 U$ has zero in the element indexed as $_{31}$ — the c' element of $U_1 U$:

$$U_2 = \begin{bmatrix} \frac{a'^*}{\sqrt{|a'|^2+|c'|^2}} & 0 & \frac{c'^*}{\sqrt{|a'|^2+|c'|^2}} \\ 0 & 1 & 0 \\ \frac{c'}{\sqrt{|a'|^2+|c'|^2}} & 0 & \frac{-a'}{\sqrt{|a'|^2+|c'|^2}} \end{bmatrix}$$

$\Longrightarrow$ the first column of $U_2 \cdot U_1 U$:

$$a' \cdot \begin{bmatrix} \frac{a'^*}{\sqrt{|a'|^2+|c'|^2}} \\ 0 \\ \frac{c'}{\sqrt{|a'|^2+|c'|^2}} \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c' \cdot \begin{bmatrix} \frac{c'^*}{\sqrt{|a'|^2+|c'|^2}} \\ 0 \\ \frac{-a'}{\sqrt{|a'|^2+|c'|^2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$\Longrightarrow$ in general,

$$U_2 U_1 U = \begin{bmatrix} 1 & d'' & g'' \\ 0 & e'' & h'' \\ 0 & f'' & j'' \end{bmatrix}$$

The norm of the first row must be 1 $\Longrightarrow$

$$1 = 1 + |d''|^2 + |g''|^2 \implies d'' = g'' = 0 \qquad \Longrightarrow \qquad U_2 U_1 U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e'' & h'' \\ 0 & f'' & j'' \end{bmatrix}$$

Because $U_2 U_1 U$ is unitary $\Longrightarrow$

$$U_2 U_1 U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \hat{U} \end{bmatrix}, \qquad \dim[\hat{U}] = 2 \times 2\,, \qquad \hat{U} \cdot \hat{U}^+ = \hat{U}^+ \cdot \hat{U} = I_2$$

- We set U_3 as

$$U_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \hat{U}^+ \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e''^* & f^{*''} \\ 0 & h''^* & j''^* \end{bmatrix}$$

$\Longrightarrow$

$$U_3 \cdot U_2 U_1 U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \hat{U}^+ \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \hat{U} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \hat{U}^+ \cdot \hat{U} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & I_2 \end{bmatrix} = I_3$$

- In general, if U is $d \times d$ unitary, we need $(d-1)$ two-level matrices $U_1, \cdots, U_{d-1}$ to achieve

$$U_{d-1} \cdot U_{d-2} \cdots U_1 \cdot U = \begin{bmatrix} 1 & 0 \\ 0 & V' \end{bmatrix}, \qquad V' \cdot V'^+ = V'^+ \cdot V' = I_{d-1}$$

$\Longrightarrow$ we continue with $(d-2)$ two-level matrices

$$U_1' = \begin{bmatrix} 1 & 0 \\ 0 & V_1' \end{bmatrix}, \qquad \cdots, \qquad U_{d-2}' = \begin{bmatrix} 1 & 0 \\ 0 & V_2' \end{bmatrix}$$

to achieve

$$U_{d-2}' \cdots U_1' \cdot \begin{bmatrix} 1 & 0 \\ 0 & V' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & V_{d-2}' \cdots V_1' \cdot V' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & V'' \end{bmatrix}$$

where

$$V'' \cdot V''^+ = V''^+ \cdot V'' = I_{d-2}$$

$\Longrightarrow$ We repeat this procedure until we get $\prod U_i^{\text{various primes}} \cdot U = I_d$; we need in total:

$$\underbrace{(d-1)}_{\text{produce } V'} + \underbrace{(d-2)}_{\text{produce } V''} + \cdots \underbrace{1}_{\text{produce } V'\cdots'} = \frac{d(d-1)}{2}$$

For n-qubits, $d = 2^n$, $\Longrightarrow$

$$\frac{d(d-1)}{2} = 2^{n-1}(2^n - 1)$$

two-level unitaries are needed in general to represent U

Single qubit and CNOT gates are universal

- Suppose that U is a two-level unitary mixing n-qubit states

$$|s\rangle \equiv |s_1 \dots s_n\rangle \qquad \text{and} \qquad |t\rangle \equiv |t_1 \dots t_n\rangle$$

It is convenient to assume that for the integers corresponding to the binary representations of $s_1 \dots s_n$ and $t_1 \dots t_n$ we have $s < t$,

$$s = 2^0 \cdot s_n + 2^1 \cdot s_{n-1} + \dots + 2^{n-1} \cdot s_1 \,, \qquad t = 2^0 \cdot t_n + 2^1 \cdot t_{n-1} + \dots + 2^{n-1} \cdot t_1$$

If $|\psi\rangle$ is

$$|\psi\rangle = \alpha\,|s\rangle + \beta\,|t\rangle + |\chi\rangle \;, \qquad \langle\chi|s\rangle = \langle\chi|t\rangle = 0\,, \qquad |\alpha|^2 + |\beta|^2 \leq 1$$

the two-level U acts nontrivially only on subspace spanned by

$$\left\{ |s\rangle \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix} , |t\rangle \equiv \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$\Longrightarrow$

$$U\,|\psi\rangle = \tilde{U} \cdot \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + |\chi\rangle \;, \qquad \tilde{U}^+\tilde{U} = \tilde{U}\tilde{U}^+ = I_2$$

Explicitly, if

$$\tilde{U} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} , \qquad \Longrightarrow \qquad U\,|\psi\rangle = (a\alpha + c\beta)\,|s\rangle + (b\alpha + d\beta)\,|t\rangle + |\chi\rangle$$

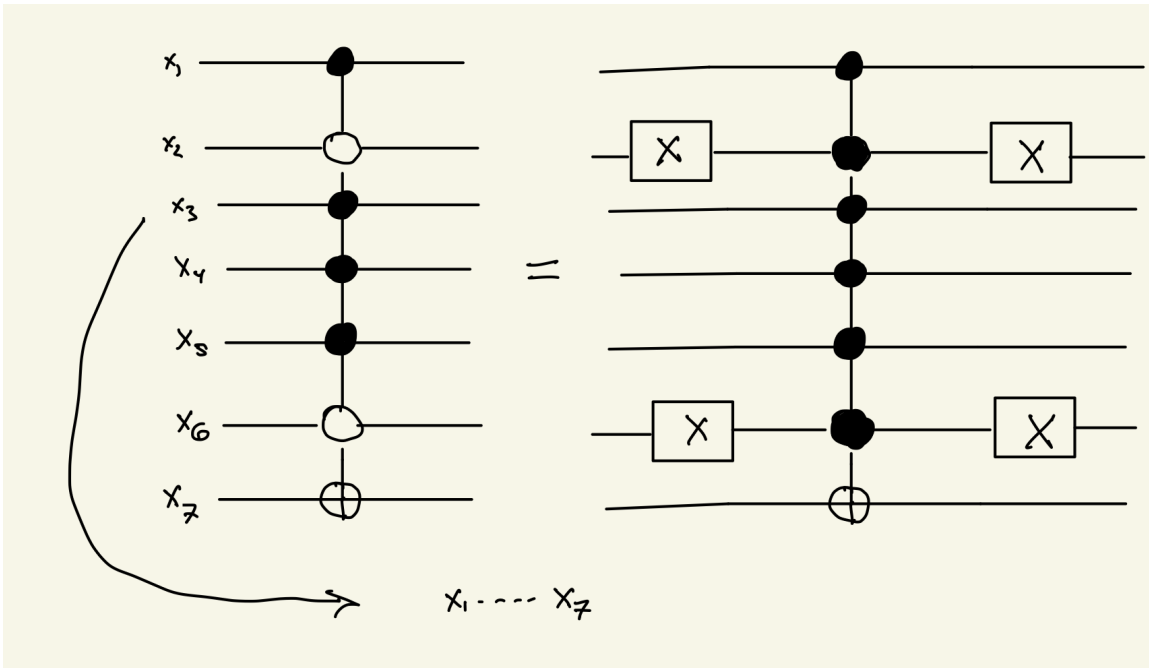
- We show that there is a quantum circuit containing only n-qubit CNOT and controlled-$\tilde{U}$ operations, that implements U.
$\Longrightarrow$ Since any of such operations can be constructed from a single qubit and a CNOT $\Longrightarrow$ we have our universality.

- The steps:
■ Construct a *Gray code* connecting binaries $s_1 \dots s_n$ and $t_1 \dots t_n$ $\Longrightarrow$ here is an example, $s = 1011101$ and $t = 1100000$:

$s =$	1	0	1	1	1	0	1
1	1	0	1	1	1	0	0
2	1	0	1	1	0	0	0
3	1	0	1	0	0	0	0
4	1	0	0	0	0	0	0
$t =$	1	1	0	0	0	0	0

i.e., , we are allowed to flip only 1 bit in each step to connect $s \to t$ binaries.

- Realize each step in the Gray code as a generalized n-qubit CNOT gate, *e.g.,* in the Gray code above, the controlled gates corresponding to steps 1 are:



- Perform controlled-$\tilde{U}$ with the first qubit as a target
- Implement identical Gray code gates in reverse order

$\Longrightarrow$ Let's see how this all works explicitly with a 2-level U acting on 3-qubit states $|s\rangle = |000\rangle$ and $|t\rangle = |111\rangle$

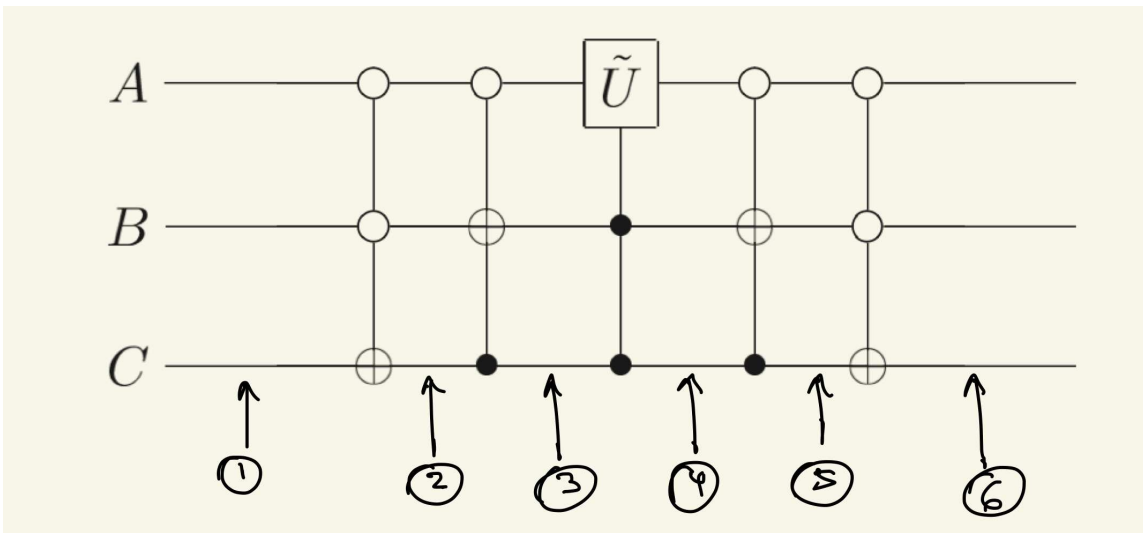
- $\dim[U] = 2^3 \times 2^3$,

$$U = \begin{bmatrix} a & 0 & 0 & 0 & 0 & 0 & 0 & c \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ b & 0 & 0 & 0 & 0 & 0 & 0 & d \end{bmatrix}, \qquad \tilde{U} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

- Gray code connecting $|s\rangle$ and $|t\rangle$:

$$\begin{array}{rccc} & x_1 & x_2 & x_3 \\ s = & 0 & 0 & 0 \\ & 0 & 0 & 1 \\ & 0 & 1 & 1 \\ t = & 1 & 1 & 1 \end{array}$$

- full quantum circuit:



- Let's show that circuit works correctly:
  - decompose

$$|\psi\rangle = \sum_{ijk} \alpha_{ijk} |x_i x_j x_k\rangle = \alpha |000\rangle + \beta |111\rangle + |\chi\rangle \ , \ \langle\chi|000\rangle = \langle\chi|111\rangle = 0$$

  - let's follow $|000\rangle$ and $|111\rangle$ subspace of $|\psi\rangle$:

$$|\psi\rangle_1 \equiv \alpha |000\rangle + \beta |111\rangle + \cdots$$

$$|\psi\rangle_2 = \alpha |001\rangle + \beta |111\rangle + \cdots$$

$$|\psi\rangle_3 = \alpha |011\rangle + \beta |111\rangle + \cdots$$

$$|\psi\rangle_4 = (a\alpha + c\beta) |011\rangle + (b\alpha + d\beta) |111\rangle + \cdots$$

$$|\psi\rangle_5 = (a\alpha + c\beta)\,|001\rangle + (b\alpha + d\beta)\,|111\rangle + \cdots$$
$$|\psi\rangle_6 = (a\alpha + c\beta)\,|000\rangle + (b\alpha + d\beta)\,|111\rangle + \cdots$$

- Note that $|\chi\rangle$ is left invariant by the circuit, *e.g.,* consider

$$|\chi\rangle_1 \equiv \alpha_{001}\,|001\rangle + \cdots$$
$$|\chi\rangle_2 = \alpha_{001}\,|000\rangle + \cdots$$
$$|\chi\rangle_3 = \alpha_{001}\,|000\rangle + \cdots$$
$$|\chi\rangle_4 = \alpha_{001}\,|000\rangle + \cdots$$
$$|\chi\rangle_5 = \alpha_{001}\,|000\rangle + \cdots$$
$$|\chi\rangle_6 = \alpha_{001}\,|001\rangle + \cdots$$

A discrete set of universal operations

$\Longrightarrow$ To complete the universality proof, we need to show that arbitrary single qubit gate,

$$U \equiv \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

can be replaced with V, such that

$$V = \prod_i V_i\,, \qquad V_i \in \left\{ T \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix},\ H \equiv \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

such that for arbitrary, no matter how small ϵ, the error of approximating U with V is smaller than ϵ:

$$E(U,V) \;\equiv\; \max_{|\psi\rangle} \;||\;(U-V)\,|\psi\rangle\;|| \qquad < \qquad \epsilon$$

$\Longrightarrow$ The general proof is technical, but here is a flavor of the proof:

- we are going to show that given circuit for a rotation $R_{\hat{n}}(\theta)$, θ **must be irrational!**, we can approximation a rotation about $\hat{n}$ by an arbitrary angle α.

  $\Longrightarrow$ Note that the rotation of the type needed can be constructed with T and H gates:

  - easy to verify that:

$$T = e^{i\frac{\pi}{8}} \cdot R_z(\pi/4) = e^{i\frac{\pi}{8}} \cdot e^{-i\frac{\pi}{8}Z}\,, \qquad HTH = e^{i\frac{\pi}{8}} \cdot e^{-i\frac{\pi}{8}X} = e^{i\frac{\pi}{8}} \cdot R_x(\pi/4)$$

  - $\Longrightarrow$

$$e^{-i\frac{\pi}{4}} \cdot T \cdot HTH = e^{-i\frac{\pi}{8}Z} \cdot e^{-i\frac{\pi}{8}X} = \cos^2\frac{\pi}{8} \cdot I - i\left(\sin\frac{\pi}{8}\cos\frac{\pi}{8} \cdot X + \sin^2\frac{\pi}{8} \cdot Y + \sin\frac{\pi}{8}\cos\frac{\pi}{8} \cdot Z\right)$$

  comparing with

$$R_{\hat{n}}(\theta) = \cos\frac{\theta}{2} \cdot I - i\sin\frac{\theta}{2}\left(n_x \cdot X + n_y \cdot Y + n_z \cdot Z\right)$$

  we identify

$$\underbrace{\cos\frac{\theta}{2} = \cos^2\frac{\pi}{8}}_{\Longrightarrow \frac{\theta}{2\pi} \text{ is irrational}}\,, \qquad \hat{n} \propto \left(\cos\frac{\pi}{8}, \sin\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$$

- *pigeonhole principle:*

  - For any arbitrary small $\delta > 0$,

$$N = \left[\frac{2\pi}{\delta}\right] + 1 \qquad N \cdot \delta > 2\pi$$

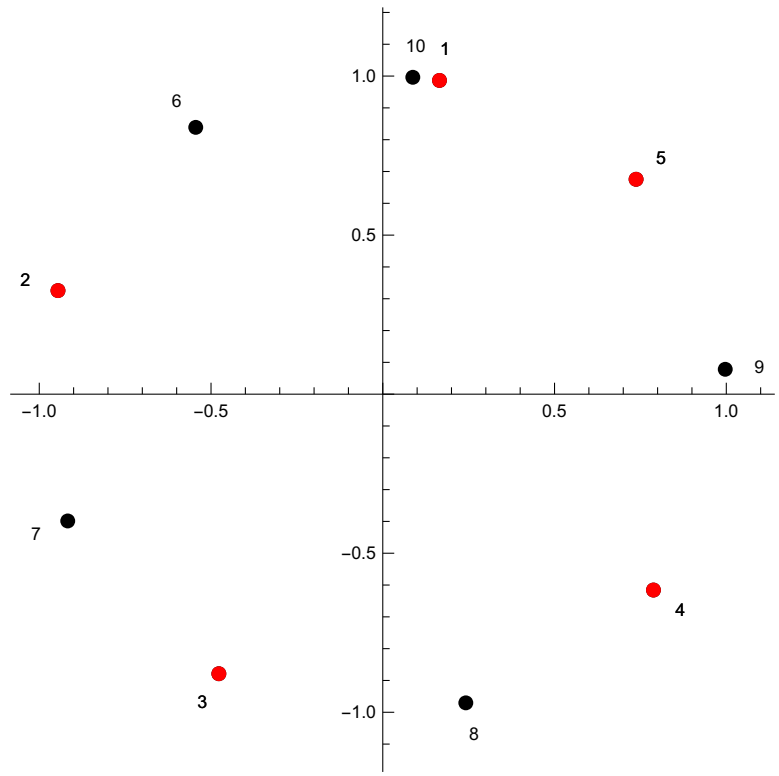
  - define set of angles θ_k, $k = 1, \cdots N$ as

$$\theta_k = k\theta (\text{mode } 2\pi)$$

  Note that

$$R_{\hat{n}}(\theta_k) = \left(R_{\hat{n}}(\theta)\right)^k$$

e.g., , with $\theta = \frac{\pi}{\sqrt{5}}$,



- red dots: $N = 5$, black+red dots: $N = 10$
- because θ is irrational, θ_k never overlap
- the smallest angle between the nearby points can not be larger than δ, since otherwise $N\delta > 2\pi$, but is larger than the sum of all the angles, which must be 2π
- for red dots, the smallest angle is $\theta_1 - \theta_5 \equiv \theta_{1-5}$
- for black+red dots, the smallest angle is $\theta_{10} - \theta_1 \equiv \theta_{10-1}$

- Let $\tilde{\theta} \equiv \theta_{i-j} < \delta$ is the smallest angle in general;

$$R_{\hat{n}}(\tilde{\theta}) = R_{\hat{n}}(\theta_i) \cdot R_{\hat{n}}(-\theta_j) = R_{\hat{n}}(\theta_i) \cdot R_{\hat{n}}(\theta_j)^+ = (R_{\hat{n}}(\theta))^i \cdot \left(R_{\hat{n}}(\theta)^+\right)^j$$

$\Longrightarrow$ rotations for $\tilde{\theta} < \delta$ can be constructed no matter how large N is, using only (by assumption existing) the $\{T, H\}$ gate construction of $R_{\hat{n}}(\theta)$

- Clearly

$$R_{\hat{n}}(\ell\tilde{\theta}) = R_{\hat{n}}(\tilde{\theta})^{\ell}$$

for integer ℓ will fill the circle with angles not larger than δ, and so can always approximate arbitrary angle α such that

$$|\ell\tilde{\theta} - \alpha| < \delta$$

- Let

$$U = R_{\hat{n}}(\alpha)\,, \qquad V = R_{\hat{n}}(\ell\tilde{\theta}) = R_{\hat{n}}(\alpha+\beta)\,, \qquad \beta < \delta$$

We now compute the error $E(U,V)$

■

$$||\,(U-V)\,|\psi\rangle\,||^2 = \langle\psi|\,(U^+-V^+)(U-V)\,|\psi\rangle = \langle\psi|\,(U^+U+V^+V-V^+U-U^+V)\,|\psi\rangle$$

$$= \langle\psi|\,(I+I-\underbrace{R_{\hat{n}}(-\alpha-\beta)}_{V^+}\,R_{\hat{n}}(\alpha)-\underbrace{R_{\hat{n}}(-\alpha)}_{U^+}\,R_{\hat{n}}(\alpha+\beta))\,|\psi\rangle = \langle\psi|\,(2I-R_{\hat{n}}(-\beta)-R_{\hat{n}}(\beta))\,|\psi\rangle$$

■

$$2I - R_{\hat{n}}(-\beta) - R_{\hat{n}}(\beta) = 2I - \left(\cos\frac{\beta}{2}\cdot I + i\sin\frac{\beta}{2}\cdot\hat{n}\cdot\vec{\sigma}\right) - \left(\cos\frac{\beta}{2}\cdot I - i\sin\frac{\beta}{2}\cdot\hat{n}\cdot\vec{\sigma}\right)$$

$$= 2I - 2I\cos\frac{\beta}{2} = 2\left(1-\cos\frac{\beta}{2}\right)\cdot I = 4\sin^2\frac{\beta}{4}\cdot I$$

$\Longrightarrow$

$$||\,(U-V)\,|\psi\rangle\,||^2 = \langle\psi|\,4\sin^2\frac{\beta}{4}\cdot I\,|\psi\rangle = 4\sin^2\frac{\beta}{4}$$

$\Longrightarrow$ (using that $\sin x < x$ for small positive x)

$$E(U,V) = \max_{|\psi\rangle}||(U-V)\,|\psi\rangle\,|| = \max_{|\psi\rangle}\left(2\sin\frac{\beta}{4}\right) = 2\sin\frac{\beta}{4} \le 2\cdot\frac{\beta}{4} = \frac{\beta}{2} < \frac{\delta}{2}$$

$\Longrightarrow$ the approximation can be made as precise as we want, if we choose small δ (large N)

- The rest of the proof is in the textbook:
  - use

$$R_{\hat{n}}(\theta) = e^{-i\frac{\pi}{4}}THTH$$

and

$$R_{\hat{m}}(\theta) \equiv H\cdot R_{\hat{n}}(\theta)\cdot H\,, \qquad \hat{m} \propto \left(\cos\frac{\pi}{8}, -\sin\frac{\pi}{8}, \cos\frac{\pi}{8}\right)\,, \qquad \hat{n}\times\hat{m} \neq \vec{0}$$

- up to a phase:

$$U = R_{\hat{n}}(\alpha) \cdot R_{\hat{m}}(\gamma) \cdot R_{\hat{n}}(\delta)$$

Approximating arbitrary gates with universal ones is hard (in general)

- Suppose we have m-gate quantum circuit operating on n-qubits, with each gate chosen from a set of g, each gate acting on f qubits.

  $\Longrightarrow$ Then the number of distinct quantum states reached by such a circuit is

$$\left((C_f^n)^g \right)^m = (C_f^n)^{g \cdot m} = \left(\frac{n!}{(n-f)!f!} \right)^{gm} \sim n^{fgm}$$

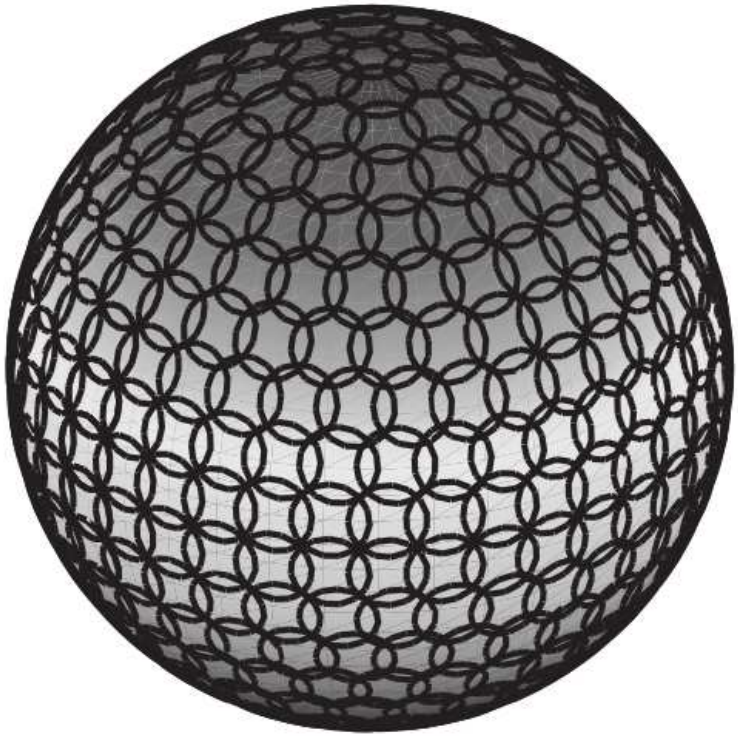
- How many patches of size ϵ n-qubit state has?

  - $$|\psi\rangle = \sum_{i_1, \cdots, i_n} \alpha_{i_1 \ldots i_n} \; |x_{i_1} \cdots x_{i_n}\rangle \qquad \Longrightarrow \qquad d = 2^n - \text{terms}$$

  - normalization:

$$\sum_{i_1, \cdots, i_n} |\alpha_{i_1 \ldots i_n}|^2 = 1 \qquad \Longrightarrow \qquad \sum_{j=1}^{d=2^n} (X_j^2 + Y_j^2) = 1\,, \qquad \alpha_{i_1 \ldots i_n} = X_j + iY_j$$

  $\Longrightarrow$ arbitrary overall phase of $|\psi\rangle$ allows to set $Y_d = 0$; thus we have set of points $\{X_j, Y_j\}$ in $\mathbb{R}^{2d-1}$ space, subject to the normalization constraint $\Longrightarrow$ $(2d-2)$-dimensional sphere S^{2d-2} (for $n = 1 \Longrightarrow$ $d = 2$ and we have our Bloch sphere $S^{2 \cdot 2 - 2} = S^2$ in $\mathbb{R}^{2 \cdot 2 - 1} = \mathbb{R}^3$)



$$\#\text{ of patches of size } \epsilon = \frac{S_{2d-2}(1)}{V_{2d-3}(\epsilon)} \sim \left(\frac{1}{\epsilon}\right)^{2d-2} = \left(\frac{1}{\epsilon}\right)^{2^{n+1}-2}$$

is the ratio of the area of n-qubit 'Bloch sphere', to the volume of the ϵ-radius patch

$\Longrightarrow$ number of gates needed must be such that

$$n^{mgf} \gtrsim \left(\frac{1}{\epsilon}\right)^{2^{n+1}-2} \qquad \Longrightarrow \qquad m \gtrsim \frac{2^n \ln \frac{1}{\epsilon}}{\ln n}$$

exponentially large in number of qubits $\Longrightarrow$ **hard!**